BirdTrack Rare Species Deduplication Pipeline

Removes duplicate sightings of the same individual bird reported by multiple observers, producing minimum/maximum individual counts per species per year. Designed for rare species with fewer than ~10 individuals nationally per year.

Quick Start

```
pip install pandas numpy scikit-learn

# Generate an example species config to edit
python species_config.py

# Run the pipeline
python dedup.py --input your_birdtrack_export.csv \
                --species species_config.csv \
                --output results/

# Run tests
python test_pipeline.py
```

Files

File	Purpose
`dedup.py`	Main pipeline — run this
`config.py`	Column name mappings + behaviour window defaults
`species_config.py`	Species config loader + example generator
`test_pipeline.py`	Validation tests with synthetic data

Setup: Column Mapping

Open `config.py` and edit `COLUMN_MAP` to match your BirdTrack CSV headers.

To inspect your headers:

```
import pandas as pd
print(pd.read_csv('your_export.csv', nrows=0).columns.tolist())
```

The values in `COLUMN_MAP` should match exactly (case-sensitive).

Setup: Species Configuration

The species config CSV controls which species are processed and how.

Generate an example to start from:

```
python species_config.py
# Creates: species_config_example.csv
```

Columns:

Column	Required	Description
`species`	Yes	Common name matching your BirdTrack records exactly
`behaviour`	Yes	`sedentary`, `mobile`, or `irruptive`
`radius_km`	No	Override default spatial radius for this species
`time_window_hours`	No	Override default time window for this species
`notes`	No	Free text for your reference

Behaviour Categories

Sedentary

A bird that remains in one location for days or weeks — rare warblers, roosting waders, wintering rarities.

- Default radius: **200m**

- Default time window: **6 hours**
- Two records at the same spot on the same morning = likely the same bird
- Same bird returning tomorrow = separate detection, but still 1 individual

Mobile

A bird moving through an area but not likely to travel far — rare waders, egrets, raptors quartering a hillside.

- Default radius: **1.5km**
- Default time window: **4 hours**
- Records at different pools within a wetland complex are flagged

Irruptive

A bird that genuinely could be anywhere — Waxwings moving between berry sources, Crossbills, autumn migrants making landfall.

- Default radius: **500m**
- Default time window: **3 hours**
- Cross-site same-day records are always flagged for manual review

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Output Files

All outputs are written to `--output` directory (default: `output/`).

`annual_counts.csv`

One row per species per year.

Column	Description
`year`	Year
`species`	Species common name
`behaviour`	Behaviour category
`min_individuals`	Conservative lower bound

`max_individuals`	Upper bound (sum of daily maxima per site)
`cluster_count`	Number of distinct detection events
`total_records`	Raw record count before deduplication

`cluster_summary.csv`

One row per detection cluster (a group of records treated as one event).

Column	Description
`cluster_id`	Internal cluster identifier
`earliest_datetime`	First record in cluster
`latest_datetime`	Last record in cluster
`sites`	Site names within cluster
`observer_count`	Number of distinct observers
`min_individuals`	Canonical count for this cluster
`flagged_for_review`	True if multiple observers involved

`review_required.csv`

All records from multi-observer clusters, with a blank `verdict` column for a human reviewer to fill in. Add verdicts:

- `same` — confirmed same individual
- `different` — confirmed different individual(s)
- `uncertain` — cannot determine

Deduplication Logic

```
1. Same visit/list ID  →  keep max count  (Tier 1)
2. Spatial clustering  →  DBSCAN with behaviour-appropriate radius
3. Temporal clustering →  within each spatial cluster, split on time gaps
4. Per cluster         →  min_individuals = max count reported
5. Per site per year   →  max_individuals = sum of daily maxima
```

```
6. Multi-observer clusters → flagged to review_required.csv
```

Tuning Tips

- **Species not found in records**: Check the exact spelling in your export. Common causes: ‘Grey Heron’ vs ‘Gray Heron’, or extra whitespace.
- **Too many clusters**: Your radius or time window is too tight. Increase `radius_km` or `time_window_hours` in the species row.
- **Too few clusters**: Windows are too generous. Tighten thresholds, or change behaviour category from `mobile` to `sedentary`.
- **No coordinates in export**: The pipeline falls back to site-name clustering. Add lat/lon to your export for better spatial resolution.
- **Lots of ‘X’ counts**: These are treated as 1. If ‘X’ means ‘flock’, consider pre-processing these records separately.

Limitations

- Without individual identification (rings, photos, song), true counts are unknowable. `min_individuals` is a defensible lower bound.
- Irruptive species on the same date at different sites may be genuinely different individuals OR the same bird relocated — only manual review of records can resolve this.
- The pipeline cannot detect the same bird returning in a later year.